



Lung Cancer

Lung cancer happens when harmful cells in your lungs grow unchecked. This can damage your lungs and spread to other parts of your body. Symptoms include persistent cough, shortness of breath and unexplained weight loss. Symptoms often don't appear until later stages. Smoking increases your risk. Screening is recommended if you're at high risk.

What Is Lung Cancer?

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Lung cancer usually starts in the airways (bronchi or bronchioles) or small air sacs (alveoli) of your lungs. It can then spread to other organs.

Lung cancer is a disease that happens when cells in your lung tissue grow out of control and make tumors. Lung cancer starts in your **airways** or the small air sacs of your lungs (alveoli). It can damage your lungs and spread to other parts of your body (metastatic lung cancer). This can eventually lead to your organs not working properly.

Lung cancer usually doesn't cause symptoms until it's in an advanced stage. That's why it's important to get screened if you're at higher risk. The most common symptoms are new or worsening cough, coughing up blood, shortness of breath, chest pain, hoarseness and unexplained weight loss.

Lung cancer is the third most common cancer in the U.S. It's the No. 1 cause of cancer-related deaths in the U.S. and worldwide. **Smoking** is the main risk factor for developing lung cancer. It's never too late to quit smoking, and quitting reduces your risk. Other risk factors include second-hand smoke, radon exposure and air pollution.

Types

Types of lung cancer include:

- **Non-small cell lung cancer (NSCLC):** NSCLC is the most common type of lung cancer. Subtypes include adenocarcinoma, squamous cell carcinoma, adenosquamous carcinoma and sarcomatoid carcinoma.
- **Small cell lung cancer (SCLC):** SCLC grows more quickly and is harder to treat than NSCLC. It's often found as a small lung tumor that's already spread to other parts of your body.

Other types of cancer can start in or around your lungs, including lymphomas (cancer in your lymph nodes), sarcomas (cancer in your bones or soft tissue) and **pleural mesothelioma** (cancer in the lining of your lungs). These are treated differently and providers don't usually call them lung cancer.

Symptoms and Causes

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Lung cancer usually starts in your airways or small air sacs (alveoli) of your lungs. It can then spread to other organs.

Symptoms of lung cancer

Symptoms could include:

- A cough that doesn't go away or gets worse over time
- Trouble breathing or **shortness of breath**
- Chest pain or discomfort
- Wheezing
- Coughing up blood
- Hoarseness
- Loss of appetite
- Unexplained weight loss
- Unexplained tiredness
- Shoulder pain
- Persistent back pain
- Persistent bone pain
- Swelling of your face, neck, arms or upper chest (**superior vena cava syndrome**)
- Small pupil and drooping eyelid in one eye, with little or no sweating on that side of your face (**Horner's syndrome**)

You won't experience every symptom. Many symptoms of lung cancer can be caused by less serious illnesses, too. Only a healthcare provider can tell you for sure what's causing these symptoms.

What are the first signs of lung cancer?

Lung cancer often doesn't cause symptoms in early stages. A cough or **pneumonia** that keeps coming back after treatment can sometimes be an early sign of lung cancer. Check with your provider if you have concerning symptoms.

Lung cancer causes

Lung cancer happens when cells keep dividing when they shouldn't. Cell division is a normal process. But all cells have a built-in "off switch" that keeps them from growing unchecked or causes them to die off when necessary. When a cell has divided too many times or has too many changes (mutations), it triggers this off switch.

Cancer cells have changes that remove the off switch. Cells keep multiplying, unchecked, and interfere with your normal cells. Cancer cells can get into your **lymph nodes** or bloodstream and move to other places in your body, spreading the damage.

Risk factors

Smoking any kind of tobacco product, including cigarettes, cigars or pipes, is the biggest single risk factor for lung cancer. Experts estimate that 80% of lung cancer deaths are smoking-related. **Vape liquids** contain substances that can harm your lungs, but it's unclear whether vaping increases your risk for cancer.

Other risk factors include:

- Being exposed to secondhand tobacco smoke
- Being exposed to harmful substances, like radon, asbestos, uranium, diesel exhaust, silica, coal products and toxins in the air
- Having previous radiation treatments to your chest (for instance, for breast cancer or lymphoma)
- Having a family history of lung cancer

Diagnosis and Tests

How doctors diagnose lung cancer

Diagnosing lung cancer can be a multistep process. Your provider will start by examining you and asking about your symptoms. They may get blood tests and a chest X-ray. Then, you may have additional tests to rule out or diagnose cancer. These tests could include:

- **Imaging:** CT scans can show more detailed pictures of changes in your lungs. A PET/CT scan can show how much cancer has spread. A brain MRI can show whether it's spread to your brain.
- **Biopsy:** If imaging suggests cancer, your provider can take samples of tissue or fluid for testing. Your provider might get a sample through **bronchoscopy**, with a needle through your skin or with surgery. A pathologist studies the sample to look for cancer cells and to determine the type of cancer.
- **Molecular testing:** A pathologist can test tissue or fluid samples for genetic changes that might affect your treatment.

Lung cancer stages

Your provider might let you know the **cancer stage** when you're diagnosed. Staging is a shorthand way to know how advanced the cancer is. It's usually based on the size of the initial tumor, how far or deep into the surrounding tissue it goes, and whether it's spread to **lymph nodes** or other organs. This helps your provider determine treatment options and affects the outlook (prognosis).

The general staging for lung cancer is:

- **Stage 0 (in situ):** There are abnormal cells in the lining of your airways.
- **Stage I:** The tumor is smaller than 3 centimeters (cm) and hasn't spread outside your lung.
- **Stage II:** The tumor is between 3 cm and 7 cm, and/or has spread to nearby lymph nodes, structures or tissues. It hasn't spread to different lobes of the same lung, to the other lung or to other organs.
- **Stage III:** The tumor is larger than 5 cm and/or has spread to nearby lymph nodes, structures or organs, but it hasn't spread to the other lung or distant organs.
- **Stage IV:** Cancer has spread to your other lung, the lining of your lungs, the fluid around your lungs or heart, the sac around your heart or distant organs.

Limited vs. extensive stage

For treatment, you might also hear SCLC (small cell lung cancer) described as limited or extensive stage. This is based on whether the area can be treated with a single radiation field.

- **Limited stage SCLC** is confined to one lung. It can sometimes be in the lymph nodes in the middle of your chest or above your collarbone on the same side.
- **Extensive stage SCLC** is widespread throughout one lung. Or it has spread to the other lung, lymph nodes on the opposite side of the lung or to other parts of your body.

Management and Treatment

How is lung cancer treated?

Treatments for lung cancer are designed to get rid of cancer in your body or slow down its growth. Your treatment will depend on the type of cancer you have, where it is, how far it's spread and many other factors. In many cases, you'll have a combination of two or more types of treatments that target cancer in different ways.

Options could include:

- **Surgery:** A surgeon removes a single tumor or sections of your lung. You might be eligible for surgery if you have a tumor that hasn't spread.
- **Chemotherapy:** Chemotherapy medications (often a combination of two or more) are designed to shrink tumors and stop cancer cells from growing.
- **Radiation therapy:** Radiation uses high-powered energy beams to destroy cancer cells.
- **Radiofrequency ablation (RFA):** RFA uses high-energy radio waves to heat and destroy cancer cells.
- **Targeted drug therapy:** These are drugs that target gene changes (like *KRAS*, *EGFR*, *HER2* and *ALK*) that help cancer grow in certain people.
- **Angiogenesis inhibitors:** These are a type of targeted drug that can keep the tumor from creating new blood vessels that cancer cells need to grow.
- **Immunotherapy:** These medications teach your immune system to fight cancer.

Some treatments can relieve symptoms instead of getting rid of the cancer. These include procedures to reduce or remove tumors that are blocking airways or to remove fluid from around your lungs.

When should I see my healthcare provider?

See a healthcare provider if you have any symptoms that concern you, especially if they've gotten worse over time. A cough that won't go away or shortness of breath are symptoms you shouldn't ignore. If you smoke or used to smoke, ask your provider about screening for lung cancer.

Outlook / Prognosis

What can I expect if I have lung cancer?

What to expect after a lung cancer diagnosis depends on many factors. If caught early, a surgeon may be able to remove the tumor. You'll need follow-up screenings for several years.

For cancer diagnosed in later stages, treatment can be a process that evolves over time. It may mean doing one type of treatment until it stops being effective, then moving on to another type.

Can lung cancer be cured?

Some types of lung cancer can be considered **cured** if diagnosed before they spread. But doctors don't often use the word "cured" to describe cancer. More common terms are "remission" or "no evidence of

disease” (NED). If you’re in **remission** or NED for five years or more, you might be considered cured. There’s always a small chance that cancer cells could come back.

Lung cancer is often diagnosed in later stages, when five-year survival rates are lowest. That’s why it’s important to see a provider for any symptoms and get screened before symptoms start if you’re at higher risk.

Prevention

Experts aren’t sure why some people develop lung cancer while others don’t. So there’s no way to completely ensure you won’t get it. But some ways to reduce your risk include:

- Don’t smoke or **quit smoking** if you do. Your risk of lung cancer starts coming down within five years of quitting.
- Avoid secondhand smoke and other substances that can harm your lungs.
- Get **screened** for lung cancer if you’re at high risk.

A note from Cleveland Clinic

A lung cancer diagnosis can bring a flood of different emotions. Sometimes, the amount of new information can be overwhelming. It’s important to remember that statistics can’t tell you how your treatment will go or what decisions are right for your specific situation. Your healthcare provider is the best resource to know what to expect.

Enlisting the help of trusted loved ones or a support group can help you consider your options and voice your preferences. Cancer treatment is often a process, and taking care of yourself is one of its most critical parts.

Experts You Can Trust

✓ **Medically Reviewed.** Last updated on 04/22/2026.

References

Cleveland Clinic’s health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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